

Electronic Signature Compliance

Legal alignment mapping for the GOYA Ledger electronic signature framework.

Covers four jurisdictions: Chile, European Union, United States, and UAE.

For detailed UAE compliance mapping, see [UAE-COMPLIANCE.md](#).

Disclaimer: This document is a **self-assessment** mapping code features to legal requirements. It is not a certification, legal opinion, or formal audit result. No third-party auditor or legal authority has validated these claims. Organizations should seek independent legal counsel before relying on these mappings for compliance purposes.

Signature levels implemented

Level	Algorithm	Biometric	Code path
Simple (FES)	Ed25519	Optional	SignatureLevel::Simple
Advanced (FEA)	ML-DSA-65 (FIPS 204)	Required (≥ 1)	SignatureLevel::Advanced
Qualified	ML-DSA-65	Required	SignatureLevel::Qualified (reserved)

Biometric evidence is stored as SHA-256 commitments only. Raw biometric data never enters the system.

Important limitation: Biometric commitments are **unverified, client-asserted** data. The system accepts whatever SHA-256 hash the client sends and does not verify that the hash corresponds to a real biometric template, a live capture, or the claimed biometric type. There is no liveness detection, no template quality check, and no trusted biometric capture device requirement. This means the biometric binding provides cryptographic integrity (the commitment is bound to the signature) but not identity assurance (anyone with the signing key can submit any hash as “biometric evidence”).

Chile — Ley 19.799

Ley 19.799 sobre Documentos Electrónicos, Firma Electrónica y Servicios de Certificación (2002, modified 2019).

Art. 2° — Definitions

Legal concept	GOYA implementation	Status
Firma electrónica (simple): cualquier sonido, símbolo o proceso electrónico que permita al receptor identificar formalmente a su autor	<code>SignatureLevel::Simple</code> — Ed25519 DID-based signature over content hash. Signer identified by <code>did:goya:{pubkey}</code>	Done
Firma electrónica avanzada : creada usando medios bajo control exclusivo del signatario, vinculada a datos de tal forma que se detecte cualquier modificación ulterior	<code>SignatureLevel::Advanced</code> — ML-DSA-65 private key (sole control) + biometric commitment bound to signing payload (tamper detection via PQC signature)	Done

Art. 3° — Legal validity

Requirement	Implementation
FES admisible como medio de prueba	Every <code>NotarizationEntry</code> with <code>Simple</code> level is stored with block height, timestamp, and signature — constitutes proof of existence
FEA equivale a firma manuscrita cuando emitida por prestador acreditado	Advanced level uses post-quantum ML-DSA-65 + biometric identity binding. Qualified TSP integration reserved for <code>SignatureLevel::Qualified</code>

Art. 5° — No discriminación

Requirement	Implementation
No se negará validez jurídica a una firma por el solo hecho de ser electrónica	Both FES and FEA produce verifiable, auditable records with cryptographic proof

Art. 15° — Certificación

Requirement	Implementation	Status
Prestador de servicios de certificación debe estar acreditado	Qualified level reserved for integration with Chilean-accredited PSC (Prestador de Servicios de Certificación)	Planned

European Union — eIDAS Regulation 910/2014

Regulation (EU) No 910/2014 on electronic identification and trust services for electronic transactions in the internal market.

Art. 3 — Definitions

eIDAS concept	GOYA implementation	Article
Electronic signature (Art. 3.10): data in electronic form attached to or associated with other electronic data, used by the signatory to sign	<code>SignatureLevel::Simple</code> — Ed25519 signature over "notarize:{signer}:{hash}"	Art. 3(10)
Advanced electronic signature (Art. 3.11): uniquely linked to signatory, capable of identifying signatory, created using data under sole control, linked to data so change is detectable	<code>SignatureLevel::Advanced</code> — see compliance table below	Art. 3(11)
Qualified electronic signature (Art. 3.12): advanced signature created by QSCD and based on qualified certificate	<code>SignatureLevel::Qualified</code> — reserved for QTSP integration	Art. 3(12)

Art. 26 — Advanced electronic signature requirements

Requirement	GOYA control	Evidence
		<code>compute_biometrics_hash()</code> bound to payload

Requirement	GOYA control	Evidence
(a) Uniquely linked to the signatory	DID (did:goya:{pubkey}) + biometric commitment(s) hashed into signing payload	
(b) Capable of identifying the signatory	Biometric evidence types: fingerprint, facial recognition, RUT, iris, voice, government ID. External identity verification supported via IdentityVerificationProvider trait (src/identity/ra.rs). RaStore::submit_and_verify() calls an external IdP and links the verified identity to the DID before allowing AdES signing. Deployment note: the SimulatedIdentityVerifier is for testing only — production deployments must plug in a real adapter (ClaveÚnica, eID, video-ident) to satisfy Art. 26(b)	BiometricType enum, BiometricEvidence.commitment, IdentityVerificationProvider trait
(c) Created using electronic signature creation data under sole control of the signatory	ML-DSA-65 private key generated client-side, never transmitted. Key encrypted with Argon2id + AES-256-GCM	Client-side key management, pqc_crypto_module
(d) Linked to the data signed so that any subsequent change is detectable	PQC signature (ML-DSA-65, 3309-byte) covers signer DID + content hash + biometric commitments hash	Signing payload: "notarize_fea: {s}:{h}:{bio_hash}"

Art. 25 — Legal effects

Provision	GOYA compliance
Art. 25(1): electronic signature not denied legal effect solely on grounds it is in electronic form	All signature levels produce verifiable records stored on-chain
Art. 25(2): qualified electronic signature has equivalent legal effect of handwritten signature	Qualified level reserved for QTSP-issued certificates

Art. 28 — Qualified certificates

Requirement	Status
Issued by qualified trust service provider (QTSP) listed in EU Trusted List	Planned — Qualified level will require external QTSP certificate chain validation

Regulation (EU) 2024/1183 — eIDAS 2.0

Enhancement	GOYA readiness
European Digital Identity Wallet (EUDI)	DID-based identity model compatible with wallet architecture
Qualified Electronic Attestation of Attributes (QEAA)	Verifiable Credentials (Credential type) with issuer DID and cryptographic signature
Post-quantum readiness recommendation	ML-DSA-65 (FIPS 204) already deployed for Advanced/Qualified levels

United States — ESIGN Act + UETA

ESIGN Act (15 U.S.C. §7001–7006)

Electronic Signatures in Global and National Commerce Act (2000).

Provision	GOYA compliance
§7001(a): signature or contract not denied legal effect solely because in electronic form	Both Simple and Advanced produce cryptographic signatures stored with audit trail
§7001(d): consumer consent requirements	Application layer responsibility (outside blockchain scope)
§7006(5) definition: “electronic, digital, or analog process attached to a contract or record, executed or adopted by a person with intent to sign”	Ed25519 and ML-DSA-65 signatures are electronic processes executed with explicit signer intent (signing payload construction)

UETA (Uniform Electronic Transactions Act)

Adopted by 49 states + DC + USVI. Provides state-level parity with ESIGN.

UETA section	GOYA compliance
§2(8): electronic signature = electronic sound, symbol, or process	Ed25519/ML-DSA-65 digital signatures qualify
§7: legal recognition of electronic records and signatures	

UETA section	GOYA compliance
	On-chain notarization with block height anchoring provides immutable record
§9: attribution — attributable to person if act of that person	DID[?]public key binding via <code>did_matches_pubkey()</code> + biometric commitment for FEA
§12: record retention satisfied by electronic record	NotarizationEntry persisted to RocksDB with content hash, timestamp, block height, signature

NIST standards alignment

Standard	Application	Status
FIPS 204 (ML-DSA)	Post-quantum signature algorithm for FEA	ML-DSA-65 (security level 3) via <code>pqc_crypto_module</code>
FIPS 186-5	Ed25519 for FES	SoftwareSigningProvider via <code>ed25519_dalek</code>
FIPS 202	SHA-3 for alias commitments	SHA3-256 via <code>pqc_crypto_module</code>
SP 800-63B	Digital identity assurance levels	FES ~ AAL1 (single factor). FEA is a structural analogy to AAL2 (cryptographic key + biometric), but not equivalent — AAL2 requires identity verification through a trusted provider and verified biometric enrollment, whereas GOYA accepts self-asserted biometric hashes without external identity proofing
SP 800-185	Vault recovery via KDF	HMAC-SHA3-256 blind index for recovery keys

Biometric evidence handling

Privacy by design

Principle	Implementation
Data minimization	Only SHA-256 commitment stored, never raw biometric data
Purpose limitation	Commitments used exclusively for signature binding

Principle	Implementation
No reversibility	SHA-256 is one-way — commitment cannot recover biometric template
Client-side processing	Biometric capture and hashing performed on client device
No central biometric database	Node stores only commitments, not templates

GDPR considerations (EU)

Article	Assessment
Art. 9 — Special categories (biometric data)	System stores SHA-256 hashes, not biometric data. Recital 51 clarifies processing must enable unique identification — one-way hashes do not. Legal basis: legitimate interest (Art. 6.1.f) or consent (Art. 6.1.a) depending on deployment
Art. 25 — Data protection by design	Commitment-only architecture prevents biometric data exposure
Art. 35 — DPIA	Recommended for deployments processing biometric commitments at scale

Chilean data protection (Ley 19.628 / Ley 21.719)

Requirement	Assessment
Datos personales sensibles (Art. 2° Ley 19.628)	SHA-256 commitments are derived data, not biometric data per se. Conservative approach: treat as sensitive and require consent
Ley 21.719 (2024) — new data protection framework	Aligns with GDPR. Same commitment-only approach applies

Endpoint coverage matrix

All endpoints that accept cryptographic signatures support both FES and FEA:

Endpoint	Simple payload	Advanced payload
POST /notarize	notarize:{s}:{h}	notarize_fea:{s}:{h}:{bio}
POST /notarize/{h}/transfer	transfer_doc:{h}:{f}:{t} vote:{id}:{opt}:{pk}	transfer_fea:{h}:{f}:{t}:{bio}

Endpoint	Simple payload	Advanced payload
POST /governance/vote/{id}		vote_fea:{id}:{opt}:{pk}:{bio}
POST /inference/submit	inference:submit:{m}:{o}	inference:submit:{m}:{o}:{bio}
POST /inference/submit-proven	inference:submit:{m}:{o}	inference:submit:{m}:{o}:{bio}
POST /inference/challenge	challenge:{id}:{h}	challenge:{id}:{h}:{bio}
POST /alias/register	alias:register:{c}	alias:register:{c}:{bio}
POST /alias/revoke	alias:revoke:{c}	alias:revoke:{c}:{bio}
POST /governance/invitations	{to}:{ids}	{to}:{ids}:{bio}
POST /governance/invitations/respond	{id}:{accepted}	{id}:{accepted}:{bio}
POST /identity/verify-signature	auto-detect Ed25519/ML-DSA-65	auto-detect

Where {bio} = SHA-256(sorted biometric commitments joined by ':').

Roadmap

Item	Status	Target
Simple (FES) — Ed25519	Done	—
Advanced (FEA) — ML-DSA-65 + biometric	Done	—
Qualified — QTSP certificate chain validation	Planned	When certified TSP available
Timestamping Authority (TSA) integration — RFC 3161	Done	—
Long-term signature preservation — CAdES/XAdES/PAdES	Done	—
eIDAS 2.0 EUDI Wallet interop	Planned	EU Digital Identity Wallet compatibility
Chilean PSC (Prestador de Servicios de Certificación) integration	Planned	Accredited provider for FEA certification

Item	Status	Target
UAE TDRA QTSP licensing	Planned	When applying for TDRA trust service provider
UAE Pass identity verification (live API)	Planned	Real-time Emirates ID verification
UAE Emirates ID validation	Done	<code>validate_emirates_id()</code> in RA module
Multi-jurisdiction RA (Chile/UAE/EU)	Done	Jurisdiction enum + <code>validate_national_id()</code>